
PRACTICE EXAM 1

NAME: SOLUTIONS

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- In this exam, the use of determinants is not allowed. We will allow determinants once we cover them in this course.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Solve the problems below concerning linear combinations.

- (a) Suppose $\mathbf{v}$, $\mathbf{w}_1$, $\mathbf{w}_2$, $\dots$, $\mathbf{w}_r$ are vectors in $\mathbb{R}^n$. Define precisely what it means for $\mathbf{v}$ to be a linear combination of the vectors $\mathbf{w}_1, \dots, \mathbf{w}_r$.

We say that $\vec{v}$ is a linear combination of $\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n$ if there exist (real) numbers $a_1, a_2, \dots, a_n$ such that

$$\vec{v} = a_1 \vec{w}_1 + a_2 \vec{w}_2 + \dots + a_n \vec{w}_n$$

- (b) Suppose m is a real number, and consider the following vectors in $\mathbb{R}^3$:

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 2 \\ m \\ m+2 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} 0 \\ 3 \\ m+1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} m \\ 1 \\ m+1 \end{bmatrix}.$$

Compute all the values of m for which $\mathbf{v}$ is a linear combination of $\mathbf{w}_1$, $\mathbf{w}_2$ and $\mathbf{w}_3$.

$\vec{v}$ is a linear combination of $\vec{w}_1, \vec{w}_2$ and $\vec{w}_3$ if and only if there exist numbers x, y, z with $\vec{v} = x\vec{w}_1 + y\vec{w}_2 + z\vec{w}_3$, i.e.,

$$\begin{bmatrix} m \\ 1 \\ m+1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ m \\ m+2 \end{bmatrix} + z \begin{bmatrix} 0 \\ 3 \\ m+1 \end{bmatrix} \Rightarrow \begin{cases} x + 2y = m \\ my + 3z = 1 \\ x + (m+1)y + (m+1)z = m+1 \end{cases} \quad (\text{i.e. } \vec{v} \text{ is a linear combination of the } \vec{w}_j \text{ if and only if this system is consistent}).$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & m \\ 0 & m & 3 & 1 \\ 1 & m+2 & m+1 & m+1 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & m \\ 0 & m & 3 & 1 \\ 0 & m & m+1 & 1 \end{array} \right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & m \\ 0 & m & 3 & 1 \\ 0 & 0 & m-2 & 0 \end{array} \right] \quad (m \text{ or } m-2 \text{ could be zero})$$

• If $m=0$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & -2 & 0 \end{array} \right] \xrightarrow{\frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & -2 & 0 \end{array} \right] \xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{2}{3} \end{array} \right] \quad \text{INCONSISTENT} \\ (\text{An equation is } 0 = \frac{2}{3})$$

• If $m=2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & 1 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{CONSISTENT}$$

• If $m \neq 0$ and $m \neq 2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & m \\ 0 & m & 3 & 1 \\ 0 & 0 & m-2 & 0 \end{array} \right] \xrightarrow{\frac{1}{m}R_2, \frac{1}{m-2}R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & m \\ 0 & 1 & \frac{3}{m} & \frac{1}{m} \\ 0 & 0 & 1 & 0 \end{array} \right] \quad \text{CONSISTENT}$$

CONCLUSION: $\vec{v}$ is a linear combination of the $\vec{w}_j$'s if and only if $m \neq 0$

(c) Use your answers from part (b) to determine the values of m for which the system

$$\begin{cases} x + 2y = m \\ my + 3z = 1 \\ x + (m+2)y + (m+1)z = m+1 \end{cases}$$

is consistent (you are **not** required to calculate the solutions to the system).

We already analyzed this exact system on part (b):

The system is consistent if and only if $m \neq 0$

Problem 2.

- (a) Suppose A is a 4×5 matrix and B is a 5×4 matrix. Find, in terms of A and B , the matrix X that satisfies

$$3X + A + B^T = 5(X^T - B)^T - 7X.$$

What is the size of X ?

$$3X + A + B^T = 5(X^T - B)^T - 7X = 5X - 5B^T - 7X = -5B^T - 2X \Rightarrow 5X = -A - 6B^T \Rightarrow X = \frac{1}{5}(-A - 6B^T)$$

X is a 4×5 matrix because both A and B^T have size 4×5 and X is the result of adding scalar multiples of A and B^T .

- (b) Find the inverse A^{-1} of the matrix

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -1 & 2 \\ 0 & 5 & -3 \end{bmatrix}.$$

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 3 & 4 & 1 & 1 & 0 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 5 & -3 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 5 & -3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 0 & -11 & 4 & -2 & 3 & 0 \\ 0 & 5 & -3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 0 & -1 & -2 & -2 & 3 & 2 \\ 0 & 5 & -3 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{-R_2} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 2 & -3 & -2 \\ 0 & 5 & -3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 - 5R_2} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 2 & -3 & -2 \\ 0 & 0 & -13 & -10 & 15 & 11 \end{array} \right] \xrightarrow{\frac{1}{-13}R_3} \left[\begin{array}{ccc|ccc} 1 & 5 & -1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 2 & -3 & -2 \\ 0 & 0 & 1 & \frac{10}{13} & -\frac{15}{13} & -\frac{11}{13} \end{array} \right] \\ &\xrightarrow{\begin{array}{l} R_1 + R_3 \\ R_2 - 2R_3 \end{array}} \left[\begin{array}{ccc|ccc} 1 & 5 & 0 & \frac{23}{13} & -\frac{29}{13} & -\frac{9}{13} \\ 0 & 1 & 0 & \frac{6}{13} & -\frac{9}{13} & -\frac{4}{13} \\ 0 & 0 & 1 & \frac{10}{13} & -\frac{15}{13} & -\frac{11}{13} \end{array} \right] \xrightarrow{R_2 - 5R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{7}{13} & \frac{17}{13} & \frac{9}{13} \\ 0 & 1 & 0 & \frac{6}{13} & -\frac{9}{13} & -\frac{4}{13} \\ 0 & 0 & 1 & \frac{10}{13} & -\frac{15}{13} & -\frac{11}{13} \end{array} \right] \quad \text{(Reduced row echelon form)} \end{aligned}$$

Thus,

$$A^{-1} = \begin{bmatrix} -\frac{7}{13} & \frac{17}{13} & \frac{9}{13} \\ \frac{6}{13} & -\frac{9}{13} & -\frac{4}{13} \\ \frac{10}{13} & -\frac{15}{13} & -\frac{11}{13} \end{bmatrix} = \frac{1}{13} \begin{bmatrix} -7 & 17 & 9 \\ 6 & -9 & -4 \\ 10 & -15 & -11 \end{bmatrix}$$

Problem 3. Determine if each of the following statements is true or false. You do not need to justify your answer.

- (a) Let A be an $n \times n$ matrix such that $A^3 = 0$. Then $I_n - A$ is invertible with inverse $I_n + A + A^2$.

TRUE $\left(\begin{array}{l} (I-A)(I+A+A^2) = I+A+A^2-A-A^2-A^3 = I \\ (I+A+A^2)(I-A) = I+A+A^2-A-A^2-A^3 = I \end{array} \right) \Rightarrow (I-A)^{-1} = I+A+A^2$

- (b) If A and B are square matrices of the same size, then $(A+B)(A-B) = A^2 - B^2$.

FALSE $\left((A+B)(A-B) = A^2 - AB + BA - B^2 \right.$, so $(A+B)(A-B) = A^2 - B^2$ only if $AB = BA$,
 $\left. \text{which is false in general} \right)$

- (c) If the system of linear equations $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the reduced row-echelon form of A has at least one row of zeros.

TRUE $\left(\begin{array}{l} \text{The reduced row-echelon form of } [A|\vec{b}] \text{ is } [R|\vec{c}], \text{ with } R \text{ the reduced row-echelon} \\ \text{form of } A. \text{ If the system is inconsistent, } [R|\vec{c}] \text{ has a row of the form } [0 \dots 0 | 1], \text{ so} \\ R \text{ has a row of zeros.} \end{array} \right)$

- (d) If a homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ has 200 equations and 40 variables, then its general solution can contain up to 160 parameters.

FALSE $\left(\begin{array}{l} \# \text{ of parameters} = (\# \text{ of variables}) - \text{Rank}(A) = 40 - \text{Rank}(A), \text{ so the solution} \\ \text{has at most 40 parameters} \end{array} \right)$

- (e) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^3 \\ y \end{bmatrix}$$

is a matrix transformation.

FALSE

$\left(\begin{array}{l} \text{If } T = T_A \text{ for some matrix } A, \text{ then } A = [T(\vec{e}_1) \ T(\vec{e}_2)] = [T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) \ T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \text{But } T_A\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}, \text{ so } T \neq T_A \text{ and } T \text{ is not a matrix transformation.} \end{array} \right)$

Problem 4.

(a) Find the general solution and a set of basic solutions to the homogeneous linear system

$$\begin{cases} 3x_1 - 6x_2 + x_3 = 0 \\ 2x_1 - 4x_2 + x_4 = 0 \end{cases}$$

$$\begin{aligned} \left[\begin{array}{cccc|c} 3 & -6 & 1 & 0 & 0 \\ 2 & -4 & 0 & 1 & 0 \end{array} \right] &\xrightarrow{R_1 - R_2} \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 0 \\ 2 & -4 & 0 & 1 & 0 \end{array} \right] &\xrightarrow{R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & -2 & 3 & 0 \end{array} \right] &\xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{cccc|c} 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\frac{3}{2} & 0 \end{array} \right] \\ &\xrightarrow{R_1 - R_2} \left[\begin{array}{cccc|c} 1 & -2 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{3}{2} & 0 \end{array} \right] \end{aligned}$$

$$\begin{cases} x_1 - 2x_2 + \frac{1}{2}x_4 = 0 \\ x_3 - \frac{3}{2}x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 2s - \frac{1}{2}t \\ x_2 = s \\ x_3 = \frac{3}{2}t \\ x_4 = t \end{cases} \quad (s, t \text{ arbitrary})$$

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The general solution is

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s - \frac{1}{2}t \\ s \\ \frac{3}{2}t \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix} \quad (s, t \text{ arbitrary})$$

A set of basic solutions is

$$\vec{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

(NOTE: $\vec{x}_1$ and $2\vec{x}_2$ also give a set of basic solutions)

(b) **Without doing any calculations**, give one solution to the system

$$\begin{cases} 3x_1 - 6x_2 + x_3 &= 7 \\ 2x_1 - 4x_2 &+ x_4 = -5 \end{cases}$$

For example:

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 7 \\ -5 \end{bmatrix}$$

(c) Use parts (a) and (b) to describe the general solution to the system

$$\begin{cases} 3x_1 - 6x_2 + x_3 &= 7 \\ 2x_1 - 4x_2 &+ x_4 = -5 \end{cases}$$

The general solution to this system is the sum of a particular solution and the general solution of the associated homogeneous system.

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 7 \\ -5 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix} \quad (s, t \text{ arbitrary})$$

Scratch working page

Scratch working page